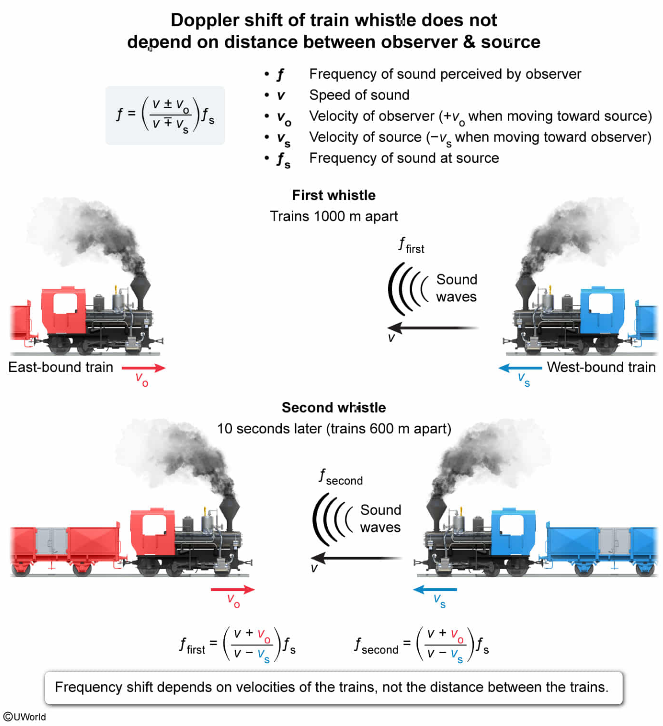
A train is traveling west at a velocity of 25 m/s. Another train is traveling east directly toward the west-bound train at a velocity of 15 m/s. The west-bound train blows its whistle with a frequency of 600 Hz when the two trains are 1000 m apart and then blows its whistle again 10 seconds later. For passengers on the east-bound train, how will the perceived frequency of the first whistle compare with the perceived frequency of the second whistle? (Note: Use 350 m/s for the speed of sound in air.)

- A. It will be 60% higher than the frequency of the second whistle. [11%]
- B. It will be 40% higher than the frequency of the second whistle. [31%]
- C. It will be approximately identical to the frequency of the second whistle. [24%]
- D. It will be 40% lower than the frequency of the second whistle. [32%]

Correct
24% answered correctly

Explanation:



The doppler effect causes the frequency of a sound emitted from a source to be perceived as a different frequency by an observer when the source and/or observer are moving relative to the Earth. The perceived sound frequency *f* is calculated from the **speed of sound** v, the observer velocity v_o, the source velocity v_s, and the frequency f_s of the sound transmitted by the source:

$$f = \left(\frac{v \pm v_o}{v \mp v_s}\right) f_s$$

When the observer is moving toward the source, the plus sign is used in the numerator. When the source is moving toward the observer, the minus sign is used in the denominator. The opposite signs are used when the observer is moving away from the source (minus in numerator) or when the source is moving away from the observer (plus in denominator).

In this question, the source is moving toward the observer and the observer is moving toward the source. The west-bound train blows its whistle the first time when the two trains are 1000 m apart. The value of f for the first whistle is given by:

$$f_{\text{first}} = \left(\frac{v + v_o}{v - v_s}\right) f_s$$

The second whistle occurs 10 seconds after the first whistle. In 10 seconds, the west-bound train travels 250 m and the east-bound train travels 150 m. Therefore, the distance between the trains is decreased by 400 m when the second whistle blows such that the two trains are 600 m apart. However, the velocities of the two trains have not changed and the value of f for the second whistle obeys the same formula:

$$f_{\text{second}} = \left(\frac{v + v_o}{v - v_s}\right) f_s$$

Therefore, despite the trains being closer for the second whistle compared to the first whistle, the perceived frequency of the two whistles will be approximately identical for the passengers on the east-bound train.

(Choices A, B, and D) The velocities of the trains are the same for the first and second whistles. Therefore, the perceived frequency of the whistle for passengers on the east-bound train will be the same for both situations.

Educational objective:
The Doppler frequency shift depends on the velocities of the source and observer but does not depend on the distance between them.

Time spent:0 sec
Subject:
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